

Friendly Order Types of Finite Posets and Ordinal Products

Incomparability Components, Forest Certificates, and Exact Transfinite Formulas

Exact formulas, composition laws, a compositionality obstruction,
forest certificates, and a finite-tail method

Research report

19 September 2026

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Main conclusions

For a finite poset P , let $c(P)$ count the connected components of its incomparability graph $\text{Inc}(P)$, including isolated vertices. Then

$$f(P) = |P| - c(P), \quad \text{equivalently} \quad w(M^r(P)) = \omega^{|P| - c(P)}.$$

So $f(P)$ is the rank of the graphic matroid of $\text{Inc}(P)$. The optimal sequences are classified and carry spanning-forest certificates.

For a non-chain finite Cartesian product of ordinals, and more generally for a product of infinite ordinals with an arbitrary finite poset, the friendly order type equals the natural product, except that one rightmost successor is removed when every ordinal factor is a successor and the finite factor has a greatest element.

Two four-element posets with identical (o, h, w, f) give different friendly order types after multiplication by the same two-element chain, so those four invariants do not support a general product formula.

Abstract. The compositional calculation of friendly order type is an explicitly identified research direction in work of Isa Vialard. We prove that for a finite poset the invariant is exactly the rank of the graphic matroid of the incomparability graph. The key lemma produces a maximal non-cut vertex while protecting any prescribed minimal root; it yields optimal sequences, spanning-forest certificates, and a classification of all maximum friendly subsets. From the formula we derive composition laws for ordinal sums, disjoint unions, lexicographic substitutions and Cartesian products (proved twice, by two independent routes), and an exact bivariate generating function for the distribution on naturally labelled posets. We then study finite successor tails of ordinal products: a projection argument shows that a nontrivial Cartesian product has a single non-endpoint incomparability component, and isolating the finite upper corner of an infinite ordinal product yields exact transfinite formulas rather than finite approximations. Well-ordered sums of two-element antichains realize every ordinal, even though all countably infinite instances have isomorphic incomparability graphs. The software compares the defining recursion with the formula on 101,660 naturally labelled posets through size seven.

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Reading guide

Sections 3–6 are finite and elementary; the component formula can be checked by reading only Sections 2–3, with no transfinite notation. Section 4 classifies the optimal witnesses and gives the matroid reading. Sections 7–8 contain the transfinite argument. The complete classification of ordinal boxes is in Theorem 9.1; the transfinite sum law and the realization of every ordinal are in Section 10; Section 11 counts naturally labelled posets by their friendly rank. Section 12 and the appendices explain exactly what the software verifies and what it does not verify.

Status and scope

Conventional proofs are supplied for the results developed here; the transfinite argument uses explicitly cited published theorems, and Appendix B separates imported statements from those proved below. The arguments are presented as proofs, but this is a research draft: it has not been independently refereed, formally verified, or checked by a proof assistant. A literature search located neither the component formula nor the exact transfinite formulas, but that is not a certification of priority, the elementary ingredients may be known in other terminology, and no publication-priority claim is made. The invariant itself, the multiset-width transfer, the both-limit product case and the binary ordinal-sum law are Vialard's and are not claimed as new. This report resolves the finite specialization, the ordinal-product family and a class of well-ordered sums; it does not resolve the unrestricted compositional program for arbitrary well-partial-orders, and it proves no formula for infinite connected pieces.

Provenance

This report is a merge of two independently produced research reports on the same invariant, originally delivered as the directories `cartesian-products-of-ordinals` and `incomparability-components-and-forests`. The first developed the transfinite side: the ordinal-cut projection lemma, the finite-tail and upper-tail lemmas, the mixed ordinal-finite product theorem, the complete classification of ordinal boxes, and the (o, h, w, f) counterexample. The second developed the finite side: the protected-root strengthening of the deletion lemma, the classification of maximum friendly subsets, the spanning-forest certificates and the graphic-matroid reading, the substitution and disjoint-union laws, the enumerative generating function, and the infinite graph-only obstruction. Both reports independently proved the shared core theorem $f(P) = |P| - c(P)$, and they proved it by the same argument: the same ordered-component lemma, the same maximal non-cut vertex lemma with the same two-case proof, the same component-capacity upper bound, and the same top-down assembly. That argument is therefore given exactly once below, in Section 3; the second report's protected-root strengthening of Lemma 3.4 is retained, since the classification of maximum friendly subsets needs it. Where the two reports genuinely differed, both are kept: the finite Cartesian-product formula is proved twice, once by the ordinal-cut projection route of Section 5 and once by the chain-grid spanning-subgraph route of Section 5.3.

1 The selected problem and its scope

1.1 A documented open direction

Vialard introduced the friendly order type $o_{\perp}(P)$ in the study of ordinal widths of finite multiset orderings. The conclusion of the MFCS 2023 paper asks about further compositional calculations of this invariant [1, Conclusion, p. 87:11]. The conclusion of the subsequent thesis again presents its compositional computation as a research direction [2, Conclusion, p. 109]. We write $f(P)$ instead of $o_{\perp}(P)$ throughout this report.

The broad question needs a precise interpretation. A construction may be computable from a rich structural description of its inputs without being a function of a short list of their ordinal invariants. These are different notions of compositionality. We investigate both, through the following restricted questions.

Question 1.1. Can the friendly order type of an arbitrary finite poset be recovered from a simple structural statistic, without constructing its entire friendly bad-sequence tree?

Question 1.2. Is $f(A \times B)$ determined by the two tuples

$$(o(A), h(A), w(A), f(A)) \quad \text{and} \quad (o(B), h(B), w(B), f(B))?$$

Question 1.3. Can one compute $f(P)$ exactly when P is a finite product of ordinals, or more generally a product of infinite ordinals with a finite poset?

Question 1.1 is the finite specialization of the published direction, and it is worth stating separately in the sharper form we actually answer.

Problem 1.4 (Finite specialization). Given an arbitrary finite poset P , identify a nonrecursive structural expression for its friendly order type. Use that expression to obtain composition laws and explicit optimal witnesses.

These precise questions are formulations adopted in this report; we do not attribute them verbatim to Vialard, and we do not claim that the literature separately labels the finite statement as a named conjecture. We answer the first and third positively and the second negatively. This resolves specified subproblems of the published program, not the unrestricted program for arbitrary well-partial-orders and arbitrary constructions.

1.2 What was already known

The limit-ordinal case must not be mistaken for a new result. Vialard's thesis already proves

$$f(A \times B) = o(A \times B) \quad \text{if both } o(A) \text{ and } o(B) \text{ are limit ordinals}$$

[2, Theorem 5.4.8(3)]. The binary ordinal-sum rule is also known [1, Proposition 4.1(1)]. The invariant itself, and its connection to multiset width, are due to Vialard [1, Definition 3.3, Theorem 3.4]. None of this may be represented as a discovery of this report. We use these facts for context, not as claims of novelty. Our transfinite proof rests on the published limit-saturation theorem stated in Section 2; the additional work is to control stripping and the finite successor corner exactly. The finite disjoint-union identity of Corollary 10.2 is likewise not claimed as a new special case; what is contributed there is its common structural explanation.

1.3 What this report establishes

Theorem 3.6 gives the finite formula $f(P) = |P| - c(P)$, and Corollary 4.3 reads it as the rank of the graphic matroid of the incomparability graph. Theorem 4.1 classifies every maximum friendly subset, and Proposition 4.2 attaches a spanning-forest certificate to the constructed witnesses. Theorem 5.2 identifies the incomparability components of every Cartesian product with two non-singleton factors, giving the product formula of Corollary 5.3; that formula is proved a second time, by an independent graph-theoretic route, in Section 5.3. Theorem 5.6 gives an exact formula for arbitrary nonempty finite lexicographic substitutions. Theorem 6.2 gives an explicit counterexample to Question 1.2. Theorem 8.1 settles the mixed ordinal-finite product problem, including the distinction between a finite factor with and without a greatest element. Theorem 10.3 gives the transfinite sum law, from which Corollary 10.5 realizes every ordinal as a friendly order type at ordinary antichain width two, while Proposition 10.6 shows that on infinite wpos the graph alone no longer determines the invariant. Theorem 11.1 computes the exact bivariate generating function for the distribution on naturally labelled posets.

A useful feature is that the strongest theorem is not restricted to countable ordinals. The proofs take place in ZFC and apply to arbitrary set-sized ordinals. The executable ordinal notation system is deliberately more restricted: it handles ordinals below ε_0 , which admit finite hereditary Cantor normal forms.

1.4 Literature and novelty status

The source check for this report was made on 19 September 2026. It included the official MFCS proceedings version, the author-hosted thesis, the earlier arXiv version, the author’s publication list, and exact-phrase searches for the invariant under its published and earlier terminology and for an incomparability-component formula [1, 2, 4, 5]. The prepublication arXiv version uses “maximal safe order type”; definitions and citations here follow the published friendly-order-type treatment. General infinite formulas not used in our proofs are deliberately not imported from earlier source versions.

The exact finite formula, the particular obstruction, and the complete successor-tail calculation below were not located in the sources checked. That is a limited literature-search conclusion, not a proof of priority, and not a guarantee that no later solution exists. The elementary ingredients may be known in other terminology. The appropriate status is: *results proved in this draft, with independent expert review and a fuller novelty check still desirable*. The mathematical assertions below have explicit proofs independent of any assertion of novelty; if the component formula turns out to be implicit in an earlier source, the proofs and artifacts remain useful but the novelty statement would need revision. That question is separate from the question whether the displayed theorems follow from the arguments given. The argument is not presented as a resolution of an unrelated famous conjecture, nor is computational agreement presented as a transfinite proof.

2 Definitions and proof dependencies

2.1 Well-partial-orders and residuals

A finite or infinite sequence $x_0, x_1, \dots$ in a poset P is *bad* if there is no pair $i < j$ with $x_i \leq x_j$. A *well-partial-order*, abbreviated wpo, has no infinite bad sequence. For $x \in P$, write

$$\uparrow x = \{y \in P : x \leq y\}, \quad P_{\not\leq x} = P \setminus \uparrow x.$$

For a finite sequence s , the residual is

$$P_{\not\leq s} = P \setminus \bigcup_{x \text{ occurring in } s} \uparrow x.$$

Every next term of a bad sequence belongs to the current residual. All orders on subsets in this report are induced orders.

A well-founded tree has rank given recursively by

$$\text{rk}(s) = \sup\{\text{rk}(s \frown x) + 1 : s \frown x \text{ is in the tree}\}, \quad \sup \emptyset = 0, \quad (1)$$

where the supremum of the empty set is zero. The tree rank is the rank of the empty sequence. In a finite tree it is the maximum number of edges on a root-to-leaf path. In general it need not be the supremum of the finite lengths of its branches; this distinction will be essential. These are the standard rank conventions for ordinal invariants of well-quasi-orders [6].

2.2 The friendly restriction

Two points are *incomparable*, written $x \perp y$, when neither is below the other. The *incomparability graph* $\text{Inc}(P)$ has vertex set P and an undirected edge between distinct incomparable elements. A bad sequence is *friendly* (called open-ended in the original definition) if, when each term x is selected, the current residual contains a point $y \perp x$. The point y is a friend witnessing that move. This is the definition from [2, Definition 5.3.2], published as [1, Definition 3.3], where the invariant is written $o_{\perp}(P)$.

The empty sequence is friendly. The friend is required in the residual *before* the choice is made; being incomparable with x , it survives that individual choice, but it need not survive later choices and need not ever be selected. These distinctions matter: requiring a permanent friend, or requiring only a friend in the original poset, would define different processes. Let $\text{Fr}(P)$ be the tree of friendly bad sequences, and let $f(P)$ be its rank. Equivalently,

$$f(P) = \sup_{\substack{x \in P \\ \exists y \in P (x \perp y)}} (f(P_{\not\leq x}) + 1). \quad (2)$$

For a finite poset this rank is the maximum number of legal moves. For an infinite wpo it is an ordinal rank, not an infinite play: all individual plays remain finite.

Lemma 2.1 (Basic properties). *For wpos $S \subseteq P$, with induced order,*

$$f(S) \leq f(P), \quad f(P) \leq o(P).$$

Also $f(P) = 0$ exactly when P is a chain, including the empty chain. If a legal sequence of k moves leaves residual R , then

$$f(P) \geq f(R) + k. \quad (3)$$

The addition in this formula is ordinary ordinal addition on the right. Finally, every subsequence of a friendly sequence is friendly.

Proof. A friendly sequence in S , with its friends in S , remains friendly in P : an earlier selected point removes precisely the same points of S in either ambient poset. Thus $\text{Fr}(S)$ is a subtree of $\text{Fr}(P)$. The friendly tree is a subtree of the full bad-sequence tree, which gives the second inequality.

Equation (2) has an eligible first move exactly when an incomparable pair exists. Next apply (2) successively along the specified legal sequence. Each step adds a successor on the right, so after k steps the lower bound is $f(R) + k$.

For the last assertion, deleting earlier choices from a friendly sequence only enlarges each subsequent residual, and it removes no comparison. The friends witnessing the original sequence therefore witness the subsequence as well. $\square$

For a finite nonempty poset with n elements one has the elementary bound

$$0 \leq f(P) \leq n - 1. \quad (4)$$

Indeed, all selected points are distinct, and the last selected point needs a different point still present as a friend. The task of Section 3 is to determine exactly how much stronger this bound becomes when the incomparability graph is disconnected.

2.3 The stripped poset

Define

$$\text{str}(P) = \{x \in P : \exists y \in P (x \perp y)\}.$$

It removes every point comparable with all points of P . Every selected point of a friendly sequence belongs to $\text{str}(P)$, so

$$f(P) \leq o(\text{str}(P)). \quad (5)$$

Notice that stripping can remove more than endpoints in a general poset. The fact that products behave much better is one of the main structural lemmas of this report.

2.4 Imported theorems

We explicitly isolate the external results used in the transfinite part. The maximal order type $o(P)$ is the rank of the bad-sequence tree. The maximal-linearization theorem of de Jongh and Parikh identifies it with an *attained* maximum among order types of well-order linear extensions. For wpos A, B ,

$$o(A \times B) = o(A) \otimes o(B), \quad (6)$$

where $\otimes$ is the natural product [3]. The use of an attained maximum, rather than just a supremum, matters in Lemma 7.3.

Theorem 2.2 (Published limit saturation). *If P is a wpo, $o(P)$ is a nonzero limit ordinal, and $o(\text{str}(P)) = o(P)$, then*

$$f(P) = o(P).$$

This is Vialard's Corollary 4.7 [1]. Its proof is not reproduced here. Accordingly, our transfinite proofs are complete *relative to this explicitly stated published theorem*, not a new self-contained derivation of the entire underlying theory.

A second imported result supplies a useful application. A finite multiset on P is a finitely supported function $P \rightarrow \mathbb{N}$. Let $M^r(P)$ be the set of finite multisets with the multiset ordering: for multisets μ, ν , cancel their common part; then $\mu <_r \nu$ when the remainder of ν is nonempty and every occurrence remaining in μ is strictly below some occurrence remaining in ν . A single witness

in ν may be reused for several occurrences, so the domination need not be injective. Adding equality gives $\leq_r$. This is *not* the distinct, injective multiset embedding order [1, Section 1]. Vialard proves

$$w(M^r(P)) = \omega^{f(P)} \quad (7)$$

for every wpo P [1, Theorem 3.4], restated as [2, Definition 5.3.2, Theorem 5.3.3]. Its technical proof is neither reproduced nor claimed as new here. We use (7) only for consequences, not for any main proof; all computations of its exponent below come from the self-contained finite argument.

The ordinal height and width are the ranks, under the convention (1), of the trees of strictly decreasing sequences and of sequences of pairwise incomparable elements, respectively. For finite posets they are the ordinary largest chain and antichain cardinalities. For infinite wpos, ordinal width is not simply the supremum of finite antichain sizes, and it may be a transfinite ordinal.

3 Every finite poset: a component formula

3.1 Incomparability components are ordered blocks

Write $c(P)$ for the number of connected components of $\text{Inc}(P)$ when P is finite, counting isolated vertices. Put $c(\emptyset) = 0$: the empty graph has zero components. When the graph rather than the poset is in focus we also write $c(\text{Inc}(P))$ for the same number.

Recall that the *ordinal sum* $A + B$ retains the internal orders of A and B and places every element of A below every element of B , whereas the *disjoint union* $A \sqcup B$ adds no cross comparisons at all. These two operations behave completely differently here, and the distinction is used throughout.

Lemma 3.1 (Ordered components). *Distinct connected components of $\text{Inc}(P)$ are uniformly comparable. They therefore form linearly ordered ordinal-sum blocks: for two distinct components C, D , either every point of C is below every point of D , or every point of D is below every point of C .*

Proof. Points in distinct components must be comparable, since otherwise they would be adjacent. Suppose $x \perp x'$ lie in one component and z lies in another. Opposite orientations, such as $x < z < x'$, contradict $x \perp x'$. Thus the orientation of comparison with z is constant along every incomparability edge and hence along every path. Applying the same argument to paths in the second component gives uniform comparison between the two components. Transitivity and antisymmetry of P make the resulting comparison a linear order on the set of components. $\square$

For a wpo the component order is itself a well-order: an infinite descending sequence of components, with one representative from each, would give an infinite strictly descending sequence in P . For the finite theorem we need only the elementary finite version of this observation.

In particular, every finite poset has a canonical decomposition

$$P = C_1 + C_2 + \cdots + C_c \quad (8)$$

as an ordinal sum of its incomparability components in increasing order. This is a decomposition into *induced* posets. It does not assert that the components are chains or antichains; they are arbitrary posets with connected incomparability graph. Nor is it a disjoint union: the blocks in (8) are completely comparable to one another.

3.2 The capacity of a single component

Lemma 3.2 (Component capacity). *In any friendly sequence in a finite poset, at most $|C| - 1$ points can be selected from each incomparability component C .*

Proof. A friend of a point of C must also lie in C , since an incomparability edge cannot join distinct components. Suppose a sequence selected every point of C , and consider its last choice from C . All other points of C would already have been selected, hence already removed, so no friend could remain in the residual. This contradicts the friend condition. Points removed incidentally, without being selected, only shrink the residual further and cannot repair the obstruction. $\square$

Corollary 3.3 (Component upper bound). *For every finite poset,*

$$f(P) \leq |P| - c(P).$$

Proof. Apply Lemma 3.2 to each of the $c(P)$ components and add. Equivalently, each component contains a point that is never selected. The argument distinguishes points that are *selected* from points removed incidentally; it does not confuse these two events. $\square$

The real issue is the lower bound: can these local capacities all be reached simultaneously? A generic graph deletion order does not suffice, because deleting an upward closure can remove several vertices at once. To avoid losing points that have not been selected, the next selected vertex should be maximal in the remaining poset. The next lemma supplies exactly this.

3.3 A maximal point can be deleted without disconnecting

A vertex of a graph is *non-cut* if deleting it leaves a connected graph. We apply this only to connected graphs with at least two vertices; the one-vertex graph left after a deletion counts as connected. The lemma below is stated in the strengthened form we shall need later: the vertex it produces can be kept away from any prescribed minimal element. That strengthening costs only two extra clauses in the same proof, and Theorem 4.1 depends on it.

Lemma 3.4 (Maximal non-cut vertex with a protected root). *Let P be finite with $|P| > 1$ and $\text{Inc}(P)$ connected, and let r be any minimal element of P . Then P has a maximal element $v \neq r$ such that $\text{Inc}(P \setminus \{v\})$ is connected. In particular, taking r to be an arbitrary minimal element, P has a maximal non-cut vertex.*

Proof. Choose a maximal element $u \neq r$. Such a u exists: if r were the only maximal element, finiteness would put every element below r , contradicting the minimality of r together with $|P| \geq 2$. If deleting u leaves a connected graph, take $v = u$ and stop.

Otherwise, by Lemma 3.1, the components of $P \setminus \{u\}$ can be written

$$C_1 < C_2 < \cdots < C_k, \quad k \geq 2.$$

Connectivity of $\text{Inc}(P)$ forces u to have a neighbor in every C_i : otherwise C_i would have no edge to u and none to any other C_j . Choose $a \in C_1$ with $a \perp u$.

For $z \in C_j$ with $j > 1$, we have $a < z$. The relation $z < u$ would imply $a < u$, while $u < z$ contradicts maximality of u . Thus

$$u \perp z \quad \text{for every } z \in C_2 \cup \cdots \cup C_k. \tag{9}$$

Choose a maximal element v of C_k . It is maximal in P : nothing in an earlier component can be above it, nothing in C_k is strictly above it, and u is incomparable with it. Moreover $v \neq r$: every element of C_1 lies below v , while r is minimal in P .

After deleting v , all the earlier components remain connected internally and attached to u . Every surviving vertex of C_k is directly adjacent to u by (9), whether or not deleting v disconnects the graph inside C_k . If $C_k = \{v\}$, this last requirement is vacuous. The remaining graph is connected in either case. $\square$

The order-theoretic word “maximal” is important, and it is the point at which transitivity of the order supplies more than graph connectivity alone. A connected graph always has non-cut vertices, but an arbitrary non-cut vertex need not delete only itself under the friendly residual operation: deleting an upward closure could remove several vertices at once and destroy the intended sequence length. A maximal point does delete only itself, because its principal upper set within the current component is a singleton.

Proposition 3.5 (Connected case). *For a finite nonempty poset with connected incomparability graph,*

$$f(P) = |P| - 1.$$

More strongly, for every minimal $r \in P$ there is a friendly sequence selecting precisely $P \setminus \{r\}$. At each step it selects a maximal vertex of the current induced poset and leaves that poset’s incomparability graph connected.

Proof. A singleton gives the empty sequence. For at least two elements, apply Lemma 3.4 with protected root r . The chosen v has a friend, since the graph is connected with more than one vertex. Since v is maximal, its upward closure in the current poset is $\{v\}$, so the residual is exactly $P \setminus \{v\}$, again with connected incomparability graph, and r is still minimal in it. Induction repeats the construction until only r remains, making $|P| - 1$ moves.

Optimality is (4), or equally Corollary 3.3 with $c(P) = 1$: no friendly sequence can select every point of a nonempty finite poset, since the final selected point would require an available incomparable point not already selected. $\square$

3.4 The exact formula

Theorem 3.6 (Finite friendly order type). *For every finite poset P ,*

$$f(P) = |P| - c(P). \quad (10)$$

Proof. The empty case is immediate. Write the canonical decomposition (8) as $P = C_1 + \cdots + C_c$.

The upper bound is Corollary 3.3.

For the lower bound, choose a minimal root r_i in each induced component C_i , and use Proposition 3.5 to obtain, inside C_i , a local friendly sequence of length $|C_i| - 1$ leaving only r_i in that component. Run these local sequences in the order $C_c, C_{c-1}, \dots, C_1$. A move in a higher component never removes a point of a lower component, so each component is still intact when its local sequence starts. A move in a lower component may remove surviving roots of higher components, but those roots are no longer needed: every local friend lies in the component currently being processed. The friends and maximal deletions inside the current component therefore remain valid, and the concatenated sequence is legal with length

$$\sum_{i=1}^c (|C_i| - 1) = |P| - c,$$

completing the proof. $\square$

3.5 Sharp bounds in terms of the stripped cardinality

Corollary 3.7. *Let $s = |\text{str}(P)|$ and let k be the number of non-singleton incomparability components of a finite poset P . Then*

$$f(P) = f(\text{str}(P)) = s - k. \quad (11)$$

If $s = 0$, then $f(P) = 0$. The value $s = 1$ is impossible. For every $s \geq 2$,

$$\left\lceil \frac{s}{2} \right\rceil \leq f(P) \leq s - 1, \quad (12)$$

and every integer in this interval occurs.

Proof. Stripping removes exactly the isolated vertices of the incomparability graph. Deleting one isolated vertex lowers the vertex count and the component count by one, leaving their difference unchanged, so Theorem 3.6 gives the equality $f(P) = f(\text{str}(P))$ and the value $s - k$. Note that this is an exact equality, sharper than the general upper bound (5).

Each remaining component has at least two vertices. Thus $1 \leq k \leq \lfloor s/2 \rfloor$ when $s > 0$, and the formula gives the bounds. For any allowed k , partition s into k integers at least two and take an ordinal sum of antichains of those sizes. Its stripped cardinality is s and its friendly order type is $s - k$. This realizes every value. $\square$

If instead one fixes only the cardinality and allows isolated vertices, every value occurs: for an n -element poset, each integer k with $0 \leq k \leq n - 1$ is realized by $\mathbf{n} - \mathbf{k} - \mathbf{1} + \Gamma_{k+1}$, with the empty-chain convention when $k = n - 1$. Here and below, $\mathbf{m}$ denotes an m -element chain and Γ_m an m -element antichain.

Proposition 3.8 (Exact monotonicity under augmentation). *Let P and Q be finite posets on the same support, and suppose every comparison of P is also a comparison of Q . Then*

$$f(P) - f(Q) = c(Q) - c(P) \geq 0.$$

Proof. Adding comparisons removes incomparability edges, and removing edges can only split components, never join them; hence $c(Q) \geq c(P)$. Subtracting the two instances of Theorem 3.6 gives the equality. $\square$

The statement applies to the completed transitive order. Adding one proposed comparison can force many others by transitivity, so a single new comparison need not lower the invariant by exactly one.

For example, $f(\Gamma_n) = n - 1$ for a nonempty n -element antichain, while $f(\mathbf{n}) = 0$ for a chain. An ordinal sum of five two-element antichains has ten stripped points but friendly order type five, whereas a ten-point antichain has friendly order type nine. The arrangement into ordinal-sum blocks, not simply the number of incomparable points, is decisive.

Combining Theorem 3.6 with the imported multiset identity gives

$$w(M^r(P)) = \omega^{|P| - c(P)} \quad (P \text{ finite}). \quad (13)$$

The finite graph statistic therefore determines an infinite ordinal width.

The degenerate cases of (13) are worth keeping explicitly. For $P = \emptyset$ there is exactly one multiset, the empty one, so the width is $1 = \omega^0$, not 0; the formula gives the right answer only because $f(\emptyset) = 0$ and $\omega^0 = 1$. For a nonempty finite chain the exponent is likewise zero, consistently with its multiset ordering being a chain.

4 Maximum friendly subsets and forest certificates

Theorem 3.6 computes a rank. The construction behind it does more: it identifies exactly which subsets of P can be the set of selected points of an optimal friendly sequence, and it equips each such sequence with a certificate that can be checked without rerunning the search.

For a subset $S \subseteq P$ of a finite poset, say that S is *friendly in P* if it admits a linear extension whose reverse listing is a friendly bad sequence in P . By the last clause of Lemma 2.1, all subsequences of that listing are then friendly as well. Conversely, a friendly sequence listing exactly the elements of S has a reverse listing that is a linear extension of S : a selected point is never above a point selected after it. This finite formulation agrees with the friendly-subset condition used in Vialard's alternative characterization [1, Definition 4.4].

Theorem 4.1 (Classification of maximum friendly subsets). *Let $C_1 < \dots < C_c$ be the incomparability components of a finite poset P . The friendly subsets of maximum possible size are exactly the sets*

$$P \setminus \{r_1, \dots, r_c\}, \quad r_i \in \text{Min}(C_i), \quad (14)$$

where $\text{Min}(C_i)$ is the set of minimal elements of the induced poset C_i . Their common size is $|P| - c$, and their number is

$$\prod_{i=1}^c |\text{Min}(C_i)|, \quad (15)$$

with the empty product equal to 1.

Proof. The construction in Theorem 3.6 realizes every choice of roots in (14), because Proposition 3.5 accepts an arbitrary prescribed minimal root of each component. This proves sufficiency and the asserted size.

For necessity, a maximum friendly subset must omit at least one member of each component by Lemma 3.2. Since its size is $|P| - c$, it omits exactly one, say r_i , from each C_i . Fix a sequence listing the subset and consider a component C_i . If r_i were not minimal in C_i , some selected $a \in C_i$ would satisfy $a < r_i$, so the root r_i would be removed at the choice of a . If that were not the last choice from C_i , no unselected member of C_i could remain as a friend for the last choice. If it were the last choice, the only possibly remaining other member of C_i would be r_i itself, which is comparable to a . Both possibilities violate the friend condition, so each r_i is minimal in its component. Different root choices give different complements, so multiplying the numbers of choices yields (15). $\square$

The theorem is stronger than a rank computation: it identifies every largest witness subset. It does not enumerate all permissible orderings of a given witness subset. Note also that the roots are minimal *inside their own component*, not necessarily minimal in P .

Proposition 4.2 (Spanning-forest certificate). *The construction in Theorem 3.6 can attach to every choice x a friend y such that the witness edges $\{x, y\}$ form a spanning tree in each original incomparability component. The unselected roots r_i are the roots of those trees.*

Proof. Within a component, choose as friend any neighbor remaining after deleting the selected maximal vertex, and orient the witness edge from that vertex to its friend. The endpoint is either selected later in that local sequence or is the protected root. Following oriented edges therefore strictly increases deletion time until the root is reached, so cycles are impossible. Every non-root has exactly one outgoing edge and eventually reaches the root, so the edges form a connected acyclic spanning subgraph of the component. The construction never uses a friend from a different component, so on the whole poset the witness edges form a spanning forest with one tree per component. $\square$

Corollary 4.3 (Graphic matroid rank and finite duality). *For a finite poset P , $f(P)$ is the maximum number of edges in a forest of $\text{Inc}(P)$, that is, the rank of the graphic matroid of $\text{Inc}(P)$. In particular,*

$$f(P^{\text{op}}) = f(P).$$

Proof. A tree on m vertices has $m - 1$ edges, so a maximum forest of a finite graph consists of one spanning tree per component and has $|P| - c(P)$ edges. By Theorem 3.6 this is $f(P)$. Reversing the order does not change which pairs are incomparable, hence does not change $\text{Inc}(P)$ or its graphic rank. $\square$

Two cautions close this section. Not every spanning forest of $\text{Inc}(P)$, and not every graph deletion order, is automatically a legal certificate: the theorem concerns the *value* of the graphic rank, while Proposition 4.2 supplies particular forests compatible with the residual process. And a forest root belonging to a later component may well be removed while an earlier component is being processed; it remains a root of the certificate forest without being an element of the final residual. Section 10 shows that the graphic-rank reading is exact for finite posets but fails for infinite ones.

5 Incomparability components of products

5.1 A projection obstruction to nontrivial ordinal cuts

An ordinal cut $P = L + U$ means that L, U partition P , are nonempty, and every element of L is strictly below every element of U . This is much stronger than saying that L is a downset.

Lemma 5.1 (Product cut lemma). *Let A, B be posets with $|A|, |B| \geq 2$. If*

$$A \times B = L + U,$$

then either $|L| = 1$ or $|U| = 1$.

Proof. Write A_L, A_U for the projections of L, U to A , and B_L, B_U for the projections to B . Every point of A_L is below every point of A_U ; the analogous assertion holds in B . Consequently each of $A_L \cap A_U$ and $B_L \cap B_U$ contains at most one point. Indeed, two points in such an intersection would each be below the other.

Because $A_L \cup A_U = A$ and $|A| \geq 2$, some element of A belongs to exactly one of the two projections. Suppose first that $a \in A_L \setminus A_U$. The whole fiber $\{a\} \times B$ lies in L , so $B_L = B$. Every element of the nonempty set B_U is then a greatest element of B , so B_U is a singleton. Choose $b \in B \setminus B_U$. The whole fiber $A \times \{b\}$ lies in L , so $A_L = A$. It follows similarly that A_U is a singleton consisting of a greatest element of A . Therefore $U \subseteq A_U \times B_U$ is a singleton.

If instead $a \in A_U \setminus A_L$, the order-dual argument makes L a singleton. No finiteness or well-foundedness assumption was used. $\square$

Theorem 5.2 (One non-endpoint component). *Let A, B be posets with $|A|, |B| \geq 2$. The incomparability components of $A \times B$ are exactly:*

- (i) *a singleton least element, when both factors have least elements;*
- (ii) *one connected component containing every non-endpoint point;*
- (iii) *a singleton greatest element, when both factors have greatest elements.*

The middle component is nonempty and has at least two points. Missing endpoints are simply omitted from this description.

Proof. By Lemma 3.1, the components are linearly ordered. There cannot be four distinct components: choose four in increasing order and cut after the second of these, including all components below it. Both sides then contain at least two points, contradicting Lemma 5.1.

If there are three components, cutting after the first forces the first to be a singleton, and cutting before the last forces the last to be a singleton. Since $|A \times B| \geq 4$, the middle has at least two points. If there are two components, the cut lemma says one is a singleton; the other has at least three points. If there is one component, it contains at least four points. Thus precisely one component is non-singleton, and any other components are endpoint singletons.

A singleton first component is a least element of the product; a singleton last component is a greatest element. Conversely, a least or greatest point is comparable with every point and hence is an isolated vertex of the incomparability graph. Finally a product has a least element exactly when both factors do, and the same equivalence holds for greatest elements. This proves all assertions. $\square$

The theorem applies to an arbitrary finite product with at least two non-singleton factors: group one non-singleton factor against the product of the others. Singleton factors can be removed without changing the poset.

5.2 Exact finite product formula

It is convenient to record the endpoint data per factor. For a nonempty poset P put

$$\varepsilon_{\perp}(P) = \begin{cases} 1 & \text{if } P \text{ has a least element,} \\ 0 & \text{otherwise,} \end{cases} \quad \varepsilon_{\top}(P) = \begin{cases} 1 & \text{if } P \text{ has a greatest element,} \\ 0 & \text{otherwise.} \end{cases}$$

These flags record *global* least and greatest elements, not merely minimal and maximal ones. A two-element antichain has two minimal elements and no least element, so $\varepsilon_{\perp} = 0$ there; the distinction is exactly what the counterexample of Section 6 exploits. A product has a least, respectively greatest, element precisely when every factor does, so for two factors we may write

$$\varepsilon_{\min} = \varepsilon_{\perp}(A)\varepsilon_{\perp}(B), \quad \varepsilon_{\max} = \varepsilon_{\top}(A)\varepsilon_{\top}(B).$$

Both formulations appear below; they are the same numbers.

Corollary 5.3. *Let A, B be finite posets of cardinality at least two. Then*

$$f(A \times B) = |A||B| - 1 - \varepsilon_{\min} - \varepsilon_{\max} = |A||B| - 1 - \varepsilon_{\perp}(A)\varepsilon_{\perp}(B) - \varepsilon_{\top}(A)\varepsilon_{\top}(B). \quad (16)$$

For at least two non-singleton finite factors, the same formula holds with the product cardinality and the all-factors endpoint flags.

Proof. The number of incomparability components is $1 + \varepsilon_{\min} + \varepsilon_{\max}$ by Theorem 5.2. Apply Theorem 3.6. $\square$

The size hypotheses are not decoration. If a factor is empty the product is empty; if a factor is a singleton the product is isomorphic to the other factor and may well be a chain. Neither degenerate case may be substituted into (16).

This is a genuine compositional formula after enriching the input data by two Boolean endpoint flags. Section 6 shows why some such additional information is necessary: it is not recoverable from the four numerical invariants in Question 1.2.

5.3 A second, graph-theoretic proof

The route just taken derives (16) from the ordinal-cut projection Lemma 5.1, which holds for arbitrary posets with no finiteness or well-foundedness hypothesis, and which is reused in the transfinite argument of Section 8. A second, independent proof of the finite case is worth recording because it is purely graph-theoretic and entirely self-contained: it replaces the cut lemma by an explicit spanning subgraph and two anchor vertices, so a reader who wants only the finite product formula need not import the projection argument at all. We keep both proofs; the projection route remains the main line, because it is the one the infinite theory uses.

Theorem 5.4 (Cartesian-product formula, second proof). *Let A, B be finite posets with $|A|, |B| \geq 2$. Then*

$$f(A \times B) = |A||B| - 1 - \varepsilon_{\perp}(A)\varepsilon_{\perp}(B) - \varepsilon_{\top}(A)\varepsilon_{\top}(B).$$

Equivalently, $\text{Inc}(A \times B)$ has exactly one non-singleton component, together with an isolated joint least element when both factors have least elements and an isolated joint greatest element when both have greatest elements.

Proof. Choose linear extensions of A and B , listing them as $a_0, \dots, a_{m-1}$ and $b_0, \dots, b_{n-1}$. Adding comparisons to make each list a chain can only delete incomparability edges from the product, so the incomparability graph of the chain grid $\mathbf{m} \times \mathbf{n}$ is a spanning subgraph of $\text{Inc}(A \times B)$.

In that grid, all vertices except the two corners $(0, 0)$ and $(m-1, n-1)$ form a connected incomparability graph. To see this, use the two incomparable anchors

$$u = (0, n-1), \quad v = (m-1, 0).$$

A vertex (i, j) with $i > 0$ and $j < n-1$ is incomparable to u ; a vertex with $i < m-1$ and $j > 0$ is incomparable to v . Every non-corner vertex is an anchor or satisfies at least one of these two conditions, and the anchors are adjacent to each other. Hence all non-corner vertices lie in one connected component, and therefore also in one component of the larger graph $\text{Inc}(A \times B)$.

It remains to classify the two possible corner exceptions. The lower corner (a_0, b_0) is isolated if a_0 and b_0 are both least elements, since it is then below every vertex. Otherwise, say a_0 is not least. Being first in a linear extension it is minimal, so some $a \in A$ is incomparable with a_0 , and then (a, b_0) is incomparable with the lower corner. It is a non-corner grid vertex, because $a \neq a_0$ and $b_0 \neq b_{n-1}$. Thus the lower corner joins the large component unless it is genuinely a least element. The argument for the upper corner is order-dual.

No further component can exist, so the number of components is $1 + \varepsilon_{\perp}(A)\varepsilon_{\perp}(B) + \varepsilon_{\top}(A)\varepsilon_{\top}(B)$, and Theorem 3.6 gives the formula. $\square$

5.4 Several factors and Boolean lattices

Corollary 5.5 (Several factors). *For finite $P_1, \dots, P_d$ with $d \geq 2$ and $|P_i| \geq 2$,*

$$f\left(\prod_{i=1}^d P_i\right) = \prod_{i=1}^d |P_i| - 1 - \prod_{i=1}^d \varepsilon_{\perp}(P_i) - \prod_{i=1}^d \varepsilon_{\top}(P_i).$$

In particular, for the Boolean lattice $\mathcal{B}_d$ of all subsets of a d -element set with $d \geq 2$,

$$f(\mathcal{B}_d) = 2^d - 3, \quad w(M^r(\mathcal{B}_d)) = \omega^{2^d-3}.$$

Proof. Apply Corollary 5.3 successively, grouping one factor against the product of the others; a product has a least, respectively greatest, element exactly when every factor does. For the Boolean lattice use $\mathcal{B}_d \cong 2^d$, which has both endpoints, and then (13). $\square$

5.5 Arbitrary lexicographic substitution

The product formula is one instance of a more general finite composition law. Let P be a finite base poset and, for each $p \in P$, let Q_p be a nonempty finite poset. The *lexicographic substitution* $P[Q_p]$ consists of the pairs (p, q) with $q \in Q_p$, ordered by

$$(p, q) \leq (p', q') \iff p < p' \text{ or } (p = p' \text{ and } q \leq q').$$

Fibers over incomparable base elements are completely incomparable to one another.

Theorem 5.6 (Substitution formula). *Let C be the set of components of $\text{Inc}(P)$. Then*

$$f(P[Q_p]) = \sum_{\substack{C \in \mathcal{C} \\ |C| \geq 2}} \left(\sum_{p \in C} |Q_p| - 1 \right) + \sum_{\{p\} \in \mathcal{C}} f(Q_p). \quad (17)$$

Proof. There are no incomparability edges between fibers lying over different base components, because those base elements are comparable and the substitution makes their whole fibers comparable in the same direction. Over a singleton component $\{p\}$, the base element is comparable to everything else, so the induced incomparability graph on its fiber is exactly $\text{Inc}(Q_p)$; by Theorem 3.6 its contribution to the vertex-minus-component count is $f(Q_p)$.

Now let C be a base component with at least two vertices. Each base vertex of C has a neighbor in C , and all fibers are nonempty. Any two vertices in the same fiber are joined by a path of length two through a neighboring fiber, since adjacent fibers are joined by all possible edges. A path in $\text{Inc}(P)$ joining two base vertices therefore lifts to a path joining any prescribed vertices of their fibers. Hence the whole collection of fibers over C forms a single connected component, with $\sum_{p \in C} |Q_p|$ vertices, contributing that number minus one. Summing over C and applying Theorem 3.6 proves the formula. $\square$

Corollary 5.7 (Uniform fibers). *Suppose every fiber is the same nonempty finite poset Q of size m . If P has n elements, s singleton incomparability components and t non-singleton incomparability components, then*

$$f(P[Q]) = m(n - s) - t + s f(Q).$$

Empty fibers cannot simply be substituted into (17): they may destroy paths in the base incomparability graph. When some fibers are empty, first delete their base vertices, take the induced order on the remaining base, and then apply the theorem to the surviving nonempty fibers.

6 A finite obstruction to scalar compositionality

6.1 A non-product baseline: three points already separate

Before the product-keyed obstruction, it is worth recording how little of the friendly order type the elementary invariants see even without forming any product. Recall that $\mathbf{m}$ is an m -element chain and Γ_m an m -element antichain.

Example 6.1 (The three-element distinction). Let

$$U = \mathbf{1} + \Gamma_2, \quad W = \mathbf{2} \sqcup \mathbf{1},$$

so U is a bottom element below two incomparable maxima, and W is a two-element chain disjoint from an isolated point. Both have cardinality and maximal order type 3, height 2, and ordinary width 2. But $\text{Inc}(U)$ has two components, an isolated bottom vertex and one edge, whereas $\text{Inc}(W)$ is connected: the isolated point of W is incomparable to both chain elements. Therefore

$$f(U) = 1, \quad f(W) = 2, \quad w(M^r(U)) = \omega, \quad w(M^r(W)) = \omega^2.$$

So the failure of determination by cardinality, height, and ordinary width already occurs on three elements. On at most two elements those data distinguish the possible isomorphism types, so three is minimal.

Note that U and W are not a counterexample to Question 1.2: they differ in f itself, so their four-tuples are not equal. The next subsection produces factors whose entire four-tuples agree, and whose products by the *same* factor still differ.

6.2 Two four-element factors

Let

$$D = \mathbf{1} + \Gamma_2 + \mathbf{1}, \quad E = \Gamma_2 + \mathbf{1} + \mathbf{1}, \quad K = \mathbf{2}.$$

Write the elements of D as a_0, a_1, a_2, a_3 , with a_0 least, $a_1 \perp a_2$, and a_3 greatest. In E , the points b_0, b_1 are incomparable minima, both below $b_2 < b_3$.

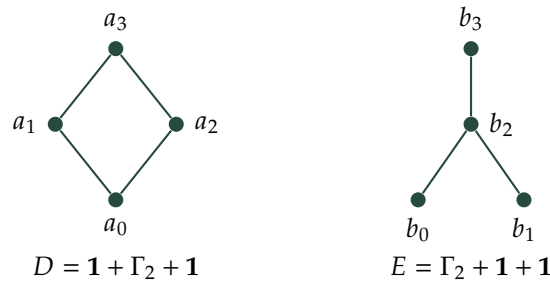


Figure 1: Hasse diagrams of the two factors. They have the same four numerical invariants but different least-element flags.

Both have four points, height three, width two, and three incomparability components. Consequently

$$(o(D), h(D), w(D), f(D)) = (o(E), h(E), w(E), f(E)) = (4, 3, 2, 1). \quad (18)$$

The factor D has both endpoints. The factor E has a greatest element but no least element. Corollary 5.3 gives

$$f(D \times K) = 8 - 3 = 5, \quad f(E \times K) = 8 - 2 = 6. \quad (19)$$

Theorem 6.2 (No formula from the four factor invariants). *There is no function Φ of eight ordinal arguments such that, for all wpos A, B ,*

$$f(A \times B) = \Phi(o(A), h(A), w(A), f(A), o(B), h(B), w(B), f(B)).$$

This fails already for finite factors of sizes four and two.

Proof. The two inputs to Φ obtained from (D, K) and (E, K) would be identical by (18). Their required outputs are different by (19). $\square$

6.3 Even the ordinary invariants of the products coincide

The products both have maximal order type eight. Each has height four. For an upper bound on height, use the rank functions

$$(0, 1, 1, 2) \text{ on } D, \quad (0, 0, 1, 2) \text{ on } E,$$

and add the K -coordinate, which is zero or one. Along a strict product comparison this sum strictly increases, so a chain has at most four points. A three-point chain in either base times K contains a four-point chain, giving equality.

Both product widths are three. In $D \times K$, an antichain is

$$\{(a_0, 1), (a_1, 0), (a_2, 0)\},$$

and a cover by three disjoint chains is

$$\begin{aligned} (a_0, 0) &< (a_1, 0) < (a_3, 0) < (a_3, 1), \\ (a_0, 1) &< (a_1, 1), \\ (a_2, 0) &< (a_2, 1). \end{aligned}$$

In $E \times K$, an antichain is

$$\{(b_0, 1), (b_1, 1), (b_2, 0)\},$$

and a three-chain cover is

$$\begin{aligned} (b_0, 0) &< (b_2, 0) < (b_3, 0) < (b_3, 1), \\ (b_0, 1) &< (b_2, 1), \\ (b_1, 0) &< (b_1, 1). \end{aligned}$$

These explicit certificates prove

$$(o, h, w)(D \times K) = (o, h, w)(E \times K) = (8, 4, 3).$$

The friendly invariant detects information absent from these ordinary product invariants as well as from the four factor invariants.

The multiset consequence is equally explicit:

$$w(M^r(D \times K)) = \omega^5, \quad w(M^r(E \times K)) = \omega^6.$$

Nothing in this example depends on numerical approximation. A complete machine-readable certificate of legal friendly sequences is included in the archive and summarized in Appendix A.

6.4 How far apart two posets with the same elementary data can be

Example 6.1 can be pushed to the largest possible separation at each cardinality.

Proposition 6.3 (Maximal separation at fixed elementary invariants). *For every $n \geq 3$, the posets*

$$L_n = \mathbf{n} - 2 + \Gamma_2, \quad R_n = \mathbf{n} - 1 \sqcup 1$$

both have cardinality and maximal order type n , height $n - 1$, and ordinary width 2, but

$$f(L_n) = 1, \quad f(R_n) = n - 1, \quad w(M^r(L_n)) = \omega, \quad w(M^r(R_n)) = \omega^{n-1}.$$

The gap $n - 2$ is the largest possible between two non-chain n -element posets.

Proof. In L_n the chain summand contributes $n - 2$ isolated vertices of the incomparability graph and the top antichain contributes one further component, so $c(L_n) = n - 1$ and $f(L_n) = 1$ by Theorem 3.6. In R_n the isolated point is incomparable to all $n - 1$ chain elements, so $\text{Inc}(R_n)$ is connected and $f(R_n) = n - 1$. The height and width assertions are immediate from the two constructions, and both maximal order types are n because both posets are finite with n elements. Every non-chain finite poset has friendly order type at least one by Lemma 2.1 and at most $n - 1$ by (4), so no larger gap is available. $\square$

This separation and the product counterexample make different points. Proposition 6.3 shows that $(|P|, o, h, w)$ leaves f almost entirely undetermined on a single poset. Theorem 6.2 shows something stronger and independent: even after f itself is added to the input data, the four-tuple still fails to determine f of a product.

7 Ordinal arithmetic and removal of finite tails

7.1 Natural arithmetic versus ordinary arithmetic

Every nonzero ordinal has a Cantor normal form

$$\alpha = \omega^{\xi_1} a_1 + \cdots + \omega^{\xi_s} a_s, \quad \xi_1 > \cdots > \xi_s, \quad 0 < a_i < \omega.$$

The exponents are arbitrary ordinals. The natural sum $\alpha \oplus \beta$ adds coefficients of equal powers after aligning the exponents. The natural product is obtained by distribution and the rule

$$(\omega^{\xi} a) \otimes (\omega^{\eta} b) = \omega^{\xi \oplus \eta} (ab).$$

For example,

$$(\omega + 2) \otimes (\omega + 3) = \omega^2 + \omega \cdot 5 + 6. \tag{20}$$

Ordinary multiplication would give a different answer. Products of posets use natural multiplication for maximal order type, by (6); ordinal sums of posets use ordinary ordinal addition.

For a successor ordinal $\gamma + 1$, define

$$\text{pred}(\gamma + 1) = \gamma.$$

We call this the *right predecessor*. We do not write a bare “ $\ominus - 1$ ” for it: ordinal subtraction conventions vary, and left subtraction would not express the operation needed here. In Cantor normal form a

successor has a positive finite constant term. Its right predecessor decreases that term by one and drops it if it becomes zero.

Every infinite ordinal has a unique decomposition

$$\alpha = \lambda + m, \quad \lambda \text{ a nonzero limit ordinal, } m < \omega. \quad (21)$$

It is a successor exactly when $m > 0$. In a natural product of finitely many ordinals, the finite constant coefficient is the product of their finite constant coefficients. In particular, a nonzero natural product has a finite constant term zero if at least one factor is a nonzero limit.

7.2 Removing finitely many points from a limit well-order

Lemma 7.1. *If Λ is a nonzero limit ordinal and $F \subseteq \Lambda$ is finite, then the induced order on $\Lambda \setminus F$ has order type Λ .*

Proof. It suffices to delete one point at a time. For a point at position β , write

$$\Lambda = \beta + 1 + \tau,$$

where τ is the order type of the interval strictly after that point. Because Λ is limit, this interval has no greatest point, so τ is a nonzero limit and in particular infinite. Hence $1 + \tau = \tau$. Deleting the point gives type $\beta + \tau = \beta + 1 + \tau = \Lambda$. Iterate this argument for the finitely many deleted points. $\square$

Lemma 7.2 (Endpoint removal). *Let P be an infinite wpo. If P has a least element b , then $o(P \setminus \{b\}) = o(P)$. If P has a greatest element g , then*

$$o(P) = o(P \setminus \{g\}) + 1.$$

Proof. Every linear extension puts a least element first and a greatest element last. Thus in the first case $o(P) = 1 + o(P \setminus \{b\})$; the ordinal on the right after the 1 is infinite, so the initial 1 is absorbed. In the second case the final 1 is not absorbed, and the displayed equality follows. Attainment of the maximal linearizations justifies both equalities directly. $\square$

The asymmetry is essential. For an infinite product, a bottom point does not cost a unit of maximal order type, while a top point does. For a finite product, both endpoints cost a point. This is why the finite and infinite formulas have different-looking corrections.

7.3 An exact finite upper-tail lemma

Lemma 7.3 (Deleting the full finite upper tail). *Suppose X is a wpo with*

$$o(X) = \Lambda + N,$$

where Λ is a nonzero limit ordinal and N is a positive integer. Let $T \subseteq X$ be an upper set with exactly N elements. Then

$$o(X \setminus T) = \Lambda. \quad (22)$$

Proof. Put $Q = X \setminus T$. Choose a maximal linear extension of X of type $\Lambda + N$. Deleting T leaves a well-order linear extension of Q . Its restriction to the first Λ positions has type Λ after finitely many deletions, by Lemma 7.1. Hence

$$o(Q) \geq \Lambda.$$

Since T is an upper set, Q is a downset. A maximal linear extension of Q , followed by any linear extension of the finite poset T , is a linear extension of X . There is no comparison from a point of T upwards into Q that could prevent this concatenation. Consequently

$$o(Q) + N \leq \Lambda + N.$$

If $o(Q) > \Lambda$, then $o(Q) \geq \Lambda + 1$ and

$$o(Q) + N \geq \Lambda + (N + 1) > \Lambda + N,$$

a contradiction. Therefore $o(Q) \leq \Lambda$, proving equality. $\square$

The equality between $|T|$ and the final finite coefficient N is part of the hypothesis. The lemma does not assert that deleting an arbitrary finite set always removes exactly the finite tail of maximal order type. Its upper-set hypothesis also has a precise role: it makes the concatenated extension of Q and T valid.

8 Products of infinite ordinals with a finite poset

8.1 Statement of the main transfinite theorem

Theorem 8.1 (Mixed ordinal–finite products). *Let $r \geq 1$, let $\alpha_1, \dots, \alpha_r$ be infinite ordinals, and let B be a nonempty finite poset. Assume*

$$r \geq 2 \quad \text{or} \quad |B| \geq 2.$$

Define

$$P = \alpha_1 \times \dots \times \alpha_r \times B, \quad \Theta = \alpha_1 \otimes \dots \otimes \alpha_r \otimes |B|.$$

Then

$$f(P) = \begin{cases} \text{pred}(\Theta), & \text{all } \alpha_i \text{ are successors and } B \text{ has a greatest element,} \\ \Theta, & \text{otherwise.} \end{cases} \quad (23)$$

In the omitted case $r = 1$ and $|B| = 1$, the poset is a chain and $f(P) = 0$.

We divide the proof into the limit case, an exact identification of the infinite core, and a finite friendly deletion of the upper corner. These are separate tasks: a maximal-order-type computation alone does not supply the required friendly witnesses.

8.2 At least one limit factor

Suppose some α_i is a limit ordinal. The natural-product formula gives $o(P) = \Theta$, and Θ is a nonzero limit because its constant Cantor coefficient is zero. There is no greatest element of P .

The nondegeneracy assumption allows application of Theorem 5.2. Thus the only possible friendless point is the least point, if B has one. Lemma 7.2 shows that its deletion does not change the infinite maximal order type. Hence

$$o(\text{str}(P)) = o(P) = \Theta.$$

Published limit saturation, Theorem 2.2, now gives $f(P) = \Theta$, as required. This argument covers the known two-limit special case and also limit factors paired with finite or successor factors in the present family.

8.3 All factors have finite successor tails

From now on, assume that every α_i is a successor. Write uniquely

$$\alpha_i = \lambda_i + m_i, \quad \lambda_i \text{ a nonzero limit}, \quad 1 \leq m_i < \omega.$$

The *finite upper corner* and its complement are

$$T = \prod_{i=1}^r [\lambda_i, \alpha_i) \times B, \quad Q = P \setminus T, \quad N = |T| = |B| \prod_{i=1}^r m_i. \quad (24)$$

The set T is upper. The constant Cantor coefficient of Θ is N , so there is a nonzero limit Λ such that

$$\Theta = \Lambda + N. \quad (25)$$

Lemma 7.3 proves $o(Q) = \Lambda$.

We next prove that Q retains enough incomparability to have friendly order type Λ . This is not automatic for an arbitrary downset of a product.

Lemma 8.2 (The core has no new friendless points). *Every point of Q other than a possible least point of P has an incomparable friend in Q . Consequently*

$$o(\text{str}(Q)) = \Lambda, \quad f(Q) = \Lambda.$$

Proof. Let $x = (x_1, \dots, x_r, b) \in Q$ be different from a possible least point. It is not a greatest point of P , since any such point belongs to T . Theorem 5.2 provides $y \in P$ with $x \perp y$. If $y \in Q$, we are done.

Otherwise $y \in T$. As $x \in Q$, choose i with $x_i < \lambda_i$. Replace just the i -th coordinate of y by $x_i + 1$ and call the resulting point y' . The limit property gives

$$x_i < x_i + 1 < \lambda_i,$$

so $y' \in Q$ and $y' \not\leq x$. Also $x \not\leq y$ was witnessed in a coordinate other than i , or in B : the original i -th coordinate of y was at least $\lambda_i > x_i$ and therefore could not have caused $x \not\leq y$. That witness is unchanged. Hence $x \not\leq y'$ and $x \perp y'$.

This shows that $\text{str}(Q)$ is either Q or Q with its least point removed. Since $o(Q) = \Lambda$ is infinite, endpoint removal leaves the same maximal order type. Applying Theorem 2.2 to Q gives $f(Q) = \Lambda$. $\square$

8.4 A finite factor with no greatest element

Assume B has no greatest element. For every $b \in B$ there is a $c \in B$ with $c \not\leq b$, because otherwise b would be greatest. For a point $x = (x_1, \dots, x_r, b) \in T$, the point

$$q_x = (0, \dots, 0, c) \in Q$$

is incomparable with x : the B -coordinate prevents $q_x \leq x$, while $x_1 \geq \lambda_1 > 0$ prevents $x \leq q_x$.

Select all N points of T in a reverse linear extension of its induced order. At each stage the selected point is maximal among the remaining points of T . Its upper set is contained in T , so the move deletes only itself. The witness q_x lies in Q , which none of these moves alters. Thus all N selections are legal and leave exactly Q . By Lemmas 2.1 and 8.2,

$$f(P) \geq f(Q) + N = \Lambda + N = \Theta.$$

The reverse inequality follows from $f(P) \leq o(P) = \Theta$. This proves the second case of (23) when all ordinal factors are successors but B has no greatest point.

8.5 A greatest point: the sharp upper bound

Now assume B has greatest element b^* . Let t_i be the greatest point of α_i , so $t_i + 1 = \alpha_i$. The product has greatest point

$$g = (t_1, \dots, t_r, b^*).$$

By Theorem 5.2, stripping removes exactly g and, possibly, the least point. Applying Lemma 7.2 gives

$$o(\text{str}(P)) = \Lambda + (N - 1) = \text{pred}(\Theta).$$

Therefore

$$f(P) \leq \Lambda + (N - 1). \quad (26)$$

If $N = 1$, the lower bound $f(P) \geq f(Q) = \Lambda$ already gives equality. No legal corner move is claimed in this case. The unique corner point is the greatest point and has no friend.

8.6 A greatest point: deleting the corner in $N - 1$ moves

Assume $N > 1$. Define the top edge

$$E = \{(\lambda_1 + j, t_2, \dots, t_r, b^*) : 0 \leq j < m_1\} \subseteq T.$$

The point g is the last point of this edge. We will select every point of T except g , but do so in an order that allows g to disappear incidentally rather than being selected illegally.

8.6.1 Phase I: the top edge below its greatest point

Select the points of $E \setminus \{g\}$ in descending order of their first coordinate. The first selection, when this phase is nonempty, removes itself and g ; each later selection in this phase removes only itself. Indeed, all coordinates except the first are already greatest, so its upper set lies on E .

The following friend survives throughout the phase. If $r \geq 2$, take

$$y = (t_1, 0, t_3, \dots, t_r, b^*) \in Q.$$

For every selected $x \in E \setminus \{g\}$, its first coordinate is strictly below t_1 , while its second coordinate is $t_2 > 0$. Thus $x \perp y$, and the second coordinate also keeps y out of all the upper sets removed in this phase.

If $r = 1$, our assumption forces $|B| \geq 2$. Choose $b_0 \neq b^*$, so $b_0 < b^*$. The point

$$y = (t_1, b_0) \in T \setminus E$$

is incomparable with each selected top-edge point and survives every Phase I deletion. This separate argument is why the single-chain exception cannot simply be ignored.

8.6.2 Phase II: the rest of the corner

Select $T \setminus E$ in any reverse linear extension. A single permanent friend works for every selection:

$$q = (0, t_2, \dots, t_r, b^*) \in Q.$$

For $x \in T \setminus E$, its first coordinate exceeds zero, so $x \not\leq q$. Since x is not on E , some other ordinal coordinate is strictly below its top value, or its B -coordinate is strictly below b^* . This prevents $q \leq x$. Hence $x \perp q$.

All upper sets of selected points remain inside T , so q survives. The reverse linear extension makes each selected point maximal among the remaining points of $T \setminus E$. All points of $E \setminus \{g\}$ have already gone. If Phase I was empty, the first Phase II selection may also delete g ; otherwise g is already absent. No unselected point of $T \setminus E$ is deleted prematurely.

There are

$$(|E| - 1) + |T \setminus E| = N - 1$$

selected points. If Phase I is empty and $N > 1$, then $T \setminus E$ is nonempty, so g is indeed deleted in Phase II. The final residual is exactly Q . It follows that

$$f(P) \geq f(Q) + (N - 1) = \Lambda + (N - 1).$$

Together with (26), this proves equality and completes the proof of Theorem 8.1. $\square$

8.7 Why the method is genuinely transfinite

The finite corner contributes a finite number of successors *after* the rank of an infinite residual. In schematic form the argument is

$$\underbrace{f(Q)}_{\Lambda} + \underbrace{\text{number of certified corner moves.}}_{N \text{ or } N-1}.$$

This is an ordinal-rank argument using the residual equation, not a claim that one can perform Λ moves before entering the corner. Every actual friendly bad sequence is finite. Transfinite complexity is encoded by the branching and nested ranks of its tree.

The core calculation and the corner strategy also have logically distinct roles. The maximal-linearization theorem and limit saturation establish $f(Q) = \Lambda$. The explicit witnesses establish the successor correction. Neither part can be replaced by a numerical extrapolation from finite grids.

9 Complete classification of ordinal boxes

9.1 All degeneracies included

Theorem 9.1 (Finite products of ordinals). *Let*

$$P = \prod_{i=1}^d \alpha_i$$

with componentwise order, where the α_i are ordinals and $d < \omega$. The empty product is a singleton. Then the following rules determine $f(P)$ completely.

- (1) *If some factor is zero, then $f(P) = 0$.*
- (2) *Otherwise discard all factors equal to one. If at most one factor remains, then $f(P) = 0$.*
- (3) *If at least two factors remain and they are all finite, then*

$$f(P) = \prod_i \alpha_i - 3.$$

(4) If at least two factors remain and at least one is infinite, put $\Theta = \bigotimes_i \alpha_i$. Then

$$f(P) = \begin{cases} \Theta, & \text{some factor is a limit ordinal,} \\ \text{pred}(\Theta), & \text{every factor is a successor ordinal.} \end{cases}$$

Proof. A zero factor makes the product empty, and singleton factors do not alter its order type. With at most one non-singleton factor the result is a chain. If every remaining factor is finite, the product has both endpoints and at least two non-singleton factors, so Corollary 5.3 gives its cardinality minus three.

In the infinite case, group the finite factors into a finite poset B ; if there are none, let B be a singleton. It has a greatest element. The nondegeneracy assumption of Theorem 8.1 holds because at least two non-singleton factors remain in total. That theorem gives exactly the last two cases. Natural-product associativity identifies its Θ with the displayed natural product of all original factors. $\square$

9.2 Worked calculations

The following examples use ordinary addition in the displayed Cantor normal forms. Multiplications by finite coefficients are written on the right.

Poset P	Maximal order type $o(P)$	Friendly order type $f(P)$
$\omega \times 3$	$\omega \cdot 3$	$\omega \cdot 3$
$(\omega + 2) \times 3$	$\omega \cdot 3 + 6$	$\omega \cdot 3 + 5$
$(\omega + 1) \times (\omega + 1)$	$\omega^2 + \omega \cdot 2 + 1$	$\omega^2 + \omega \cdot 2$
$(\omega + 2) \times (\omega + 3)$	$\omega^2 + \omega \cdot 5 + 6$	$\omega^2 + \omega \cdot 5 + 5$
$(\omega + 1) \times \Gamma_2$	$\omega \cdot 2 + 2$	$\omega \cdot 2 + 2$
$(\omega + 1) \times D$	$\omega \cdot 4 + 4$	$\omega \cdot 4 + 3$

Table 1: Exact values. The antichain example uses the mixed-product theorem rather than the ordinal-box corollary: its finite factor has no greatest element.

For an example with three nonconstant terms, distribution gives

$$(\omega^2 + 3) \otimes (\omega + 2) = \omega^3 + \omega^2 \cdot 2 + \omega \cdot 3 + 6.$$

Both factors are successors, so

$$f((\omega^2 + 3) \times (\omega + 2)) = \omega^3 + \omega^2 \cdot 2 + \omega \cdot 3 + 5.$$

For an example requiring hereditary exponents,

$$\begin{aligned} & (\omega^\omega + \omega + 4) \otimes (\omega^2 + 2) \\ &= \omega^{\omega+2} + \omega^\omega \cdot 2 + \omega^3 + \omega^2 \cdot 4 + \omega \cdot 2 + 8. \end{aligned}$$

The friendly order type of the corresponding product is the same expression with final coefficient 7 instead of 8. Here the leading exponent arises from the natural sum $\omega \oplus 2 = \omega + 2$.

There is no countability restriction. For instance,

$$f((\omega_1 + 1) \times 2) = \omega_1 \cdot 2 + 1.$$

One can calculate the natural product using $\omega_1 = \omega^{\omega_1}$: all powers ω^ξ for $\xi < \omega_1$ are countable and their supremum is ω_1 . Thus its Cantor form is a single monomial, and natural multiplication by two doubles the coefficient. This example is a mathematical consequence of the theorem, not an output of the below- ε_0 software.

9.3 Finite-grid approximation is not a proof

For finite $n \geq 2$, the finite product formula gives

$$f(n \times 2) = 2n - 3.$$

The supremum of these values is ω , whereas

$$f(\omega \times 2) = \omega \cdot 2.$$

Thus even an exhaustive list of all finite rectangles of this shape would not determine the friendly order type of their union by taking numerical suprema. A friendly-tree node can have arbitrarily high residual rank without there being a branch of corresponding infinite length. The finite checks in the archive are deliberately not advertised as a verification of the transfinite theorem.

This caution is about *method*: numerical suprema of finite instances do not compute a transfinite rank. It should not be confused with the separate, mathematical obstruction of Proposition 10.6 below, which says that on infinite wpos the isomorphism type of the incomparability graph no longer determines the invariant at all, however it is computed. The first is a warning about extrapolating from data; the second is a theorem about what the data could possibly contain.

10 Ordinal sums and a compositional class

10.1 The sum rule

The binary identity below is already known [2, Theorem 5.4.8(1)]. We give a proof to fix the direction of ordinal addition and then record its transfinite extension.

Proposition 10.1. *For wpos A, B ,*

$$f(A + B) = f(A) + f(B).$$

Proof. Use transfinite induction on $f(B)$, simultaneously for all A . A friendly first move in A must have its friend in A and deletes all of B . These first moves contribute a supremum of $f(A)$ to the residual rank formula. A friendly first move b in B has its friend in B and leaves $A + B_{\not\leq b}$. Since $f(B_{\not\leq b}) + 1 \leq f(B)$, induction applies to this residual and gives

$$f(A + B_{\not\leq b}) + 1 = f(A) + f(B_{\not\leq b}) + 1.$$

If $f(B) = 0$, there are no such moves, giving the result directly. If $f(B) > 0$, take the supremum over them. Ordinary ordinal addition is continuous in its right argument at limits, and a supremum which is a successor is attained. Consequently this supremum is

$$f(A) + \sup_b (f(B_{\not\leq b}) + 1) = f(A) + f(B).$$

It dominates the contribution $f(A)$ from first moves in A . This proves the formula in all cases, including empty factors. $\square$

The induction may equally be run on the rank of $\text{Bad}(B)$ rather than on $f(B)$: a residual $B_{\not\leq b}$ has strictly smaller bad-sequence rank, and nothing else in the argument changes. This is a choice of induction measure, not a different proof.

The finite case of the ordinal sum has a companion for the disjoint union, which behaves in the opposite way.

Corollary 10.2 (Finite sums and disjoint unions). *For finite posets A, B ,*

$$f(A + B) = f(A) + f(B).$$

If moreover A and B are both nonempty, then

$$f(A \sqcup B) = |A| + |B| - 1.$$

Proof. The first identity is the finite case of Proposition 10.1; equivalently, the incomparability graph of an ordinal sum is the disjoint union of the two incomparability graphs, so vertex and component counts both add and Theorem 3.6 applies. In the poset disjoint union, by contrast, every vertex of A is incomparable to every vertex of B , so the resulting incomparability graph is connected when both parts are nonempty. Hence $c(A \sqcup B) = 1$, and Theorem 3.6 gives $|A| + |B| - 1$. $\square$

The disjoint-union identity is consistent with the published finite formulas and is not claimed here as a new special case; what the component theorem adds is the common structural explanation of both identities.

Theorem 10.3 (Well-ordered sums). *Let $(P_i)_{i < \eta}$ be wpos indexed by an ordinal. Their ordinal sum is a wpo, and*

$$f\left(\sum_{i < \eta} P_i\right) = \sum_{i < \eta} f(P_i), \quad (27)$$

where the right-hand sum is the ordinary transfinite ordinal sum.

Proof. An infinite bad sequence in the sum cannot move to a strictly higher block. Its block indices would therefore be nonincreasing and eventually constant, since ordinals have no infinite strict descent. Its tail would be an infinite bad sequence within one block, a contradiction.

The friendly formula follows by transfinite induction on η . The successor step is Proposition 10.1. Suppose η is limit. Each initial subsum embeds into the whole sum, so monotonicity gives the lower bound by the supremum of their friendly order types. For the upper bound, a first move in block P_j deletes all higher blocks. It is legal precisely when it is legal in the subsum through P_j , since its friend must lie in the same block. Its residual and rank contribution are therefore the same as in that initial subsum. Taking the supremum of all first-move contributions gives no more than the supremum of the friendly order types of the initial subsums. This equals the transfinite sum on the right of (27). $\square$

For finite blocks the formula becomes especially concrete:

$$f\left(\sum_{i < \eta} P_i\right) = \sum_{i < \eta} (|P_i| - c(P_i)). \quad (28)$$

The finite subtraction takes place *inside each summand*; the outer sum is ordinal addition in the displayed order, not natural addition and not cardinal addition. Consequently a final finite block can survive in the answer where the same block placed at the beginning would be absorbed.

Corollary 10.4 (Arbitrary wpo with finite components). *Let P be a wpo all of whose incomparability components are finite, and let $(C_\xi)_{\xi < \eta}$ be those components in their canonical well-ordered sequence. Then*

$$f(P) = \sum_{\xi < \eta} (|C_\xi| - 1). \quad (29)$$

Proof. Lemma 3.1 exhibits P as the well-ordered sum of its components, which is a well-order by the remark following that lemma. Each block is finite with connected incomparability graph, so $|C_\xi| - c(C_\xi) = |C_\xi| - 1$. Apply (28). $\square$

10.2 Explicit transfinite values

Write $H = \sum_{n < \omega} \Gamma_2$, an ω -indexed ordinal sum of two-element antichains. Then

$$f(H) = \omega, \quad w(M^r(H)) = \omega^\omega.$$

For $k < \omega$,

$$f(H + \Gamma_{k+1}) = \omega + k, \quad w(M^r(H + \Gamma_{k+1})) = \omega^{\omega+k},$$

so the final finite block is not absorbed. Likewise $\sum_{\xi < \omega^2} \Gamma_2$ has friendly order type ω^2 . Inserting arbitrary chain blocks anywhere in such a sum adds nothing, since their contribution is zero: for instance, appending a chain of type ω to H gives $f(H + \omega) = \omega + 0 = \omega$, whereas $o(H + \omega) = \omega \cdot 2$. A long comparable tail can change maximal order type without creating any new friendly moves.

Corollary 10.5 (Every ordinal, at ordinary width two). *For every ordinal α put*

$$P_\alpha = \sum_{\xi < \alpha} \Gamma_2.$$

Then $f(P_\alpha) = \alpha$ and $w(M^r(P_\alpha)) = \omega^\alpha$. For $\alpha > 0$ the ordinary maximum antichain size of P_α is exactly two.

Proof. Every two-element antichain contributes $2 - 1 = 1$ to (27), and the transfinite sum of α copies of 1 is α . An antichain of P_α cannot meet two different blocks, since distinct blocks are completely comparable, and each nonempty block has an antichain of size two. Apply (7) for the width. $\square$

Proposition 10.6 (The infinite graph-only obstruction). *On infinite wpos, the friendly order type is not determined by the isomorphism type of the incomparability graph, even when every component is finite. Indeed, for all countably infinite ordinals α the graphs $\text{Inc}(P_\alpha)$ are pairwise isomorphic, while $f(P_\alpha) = \alpha$.*

Proof. For countably infinite α , the graph $\text{Inc}(P_\alpha)$ is a disjoint union of countably infinitely many isolated edges, one per block, and any bijection between two such block index sets induces a graph isomorphism. The friendly order types are given by Corollary 10.5 and need not coincide: ω and $\omega + 1$ already differ. $\square$

So the finite graphic-rank reading of Corollary 4.3 is exact but does not extend unchanged to infinite graphs. What the infinite case additionally requires is the *order* of the components, which carries indispensable ordinal information that the abstract graph discards. Note what is and is not claimed: these transfinite values are consequences of the sum law applied to blocks we control completely. They are not a claim that the general connected infinite case has been solved.

10.3 A usable structural language

The results provide an exact evaluator on descriptions built from finite posets, finite ordinal boxes, finite disjoint unions and lexicographic substitutions with nonempty finite fibers (Corollary 10.2 and Theorem 5.6), the mixed products of Theorem 8.1, and well-ordered sums of already evaluated expressions. On the finite part the evaluator needs no recursion at all: by Theorem 3.6 it reads the answer off the incomparability graph. A product node in this language must satisfy the proved hypotheses. An arbitrary product of two previously generated expressions is not automatically covered.

This distinction prevents an erroneous overstatement. Closure under ordinal sums, together with some exact product rules, is not closure under all Cartesian products. The counterexample in Theorem 6.2 shows that a general evaluator cannot merely store the tuple (o, h, w, f) and discard the rest of the structure.

11 Enumeration and a bivariate generating function

The component decomposition (8) is rigid enough to count finite posets by their friendly rank exactly, not merely to tabulate them.

Call a poset on $\{0, 1, \dots, n-1\}$ *naturally labelled* when $i <_P j$ implies $i < j$ as integers. Let a_n count such posets, with $a_0 = 1$, and let b_n count those of them whose incomparability graph is connected, for $n \geq 1$. These are neither counts of unlabelled posets nor counts of all labelled posets: every isomorphism type has at least one natural labelling, but it usually has many. Put

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad B(z) = \sum_{n \geq 1} b_n z^n.$$

In this section only, the sans-serif letters A and B denote formal power series, not posets; no analytic convergence is asserted. Let $a_{n,k}$ count naturally labelled n -element posets with friendly order type k .

Theorem 11.1 (Distribution by component rank). *One has*

$$A(z) = \frac{1}{1 - B(z)}, \quad b_n = a_n - \sum_{i=1}^{n-1} b_i a_{n-i}, \quad (30)$$

and the bivariate distribution is

$$\sum_{n,k \geq 0} a_{n,k} z^n u^k = \frac{1}{1 - \sum_{m \geq 1} b_m z^m u^{m-1}}. \quad (31)$$

For $n \geq 1$ and $0 \leq k \leq n-1$,

$$a_{n,k} = [z^n] B(z)^{n-k}. \quad (32)$$

Proof. In a natural labelling, if $C < D$ are incomparability components then every label in C is smaller than every label in D , by Lemma 3.1. Components therefore occupy consecutive intervals of labels in their canonical order, and (8) becomes a unique ordered list of nonempty naturally labelled posets with connected incomparability graph. Conversely, concatenating any such list by ordinal sum yields a naturally labelled poset with exactly those components. Summing over ordered lists gives $1 + B + B^2 + \dots$, which is the first identity; extracting coefficients gives the recurrence.

A component of size m contributes $m - 1$ to the friendly order type by Theorem 3.6, so weighting it by $z^m u^{m-1}$ proves (31). A poset of size n and friendly order type k has exactly $n - k$ components, so extracting from lists of that length gives (32). $\square$

The exhaustive enumeration described in Section 12 gives

$$(a_0, \dots, a_7) = (1, 1, 2, 7, 40, 357, 4824, 96428)$$

and, through (30),

$$(b_1, \dots, b_7) = (1, 1, 4, 27, 275, 4070, 86278).$$

These values are outputs of the included exact enumeration, recomputed for this report, not numbers taken from a database without verification.

Corollary 11.2 (Fixed-rank counting polynomials). *For fixed $k \geq 0$ and $n \geq k + 1$,*

$$a_{n,k} = \sum_{\substack{m_1, \dots, m_k \geq 0 \\ \sum_{j=1}^k j m_j = k}} \frac{(n-k)_r}{\prod_{j=1}^k m_j!} \prod_{j=1}^k b_{j+1}^{m_j}, \quad r = \sum_{j=1}^k m_j, \quad (33)$$

where $(x)_r = x(x-1)\cdots(x-r+1)$ and empty products are 1. As a function of n this is a polynomial of degree k with leading coefficient $1/k!$.

Proof. There are $n - k$ component positions. Let m_j record how many components have size $j + 1$; their total rank contribution is $\sum_j j m_j = k$, and all remaining components are singletons. Choose and order the r non-singleton positions in $(n-k)_r / \prod_j m_j!$ ways, then choose the connected structure on each. The degree is at most $r \leq k$, and degree k occurs only when $m_1 = k$; since $b_2 = 1$, that term is $\binom{n-k}{k}$, whose leading coefficient is $1/k!$. $\square$

For example, within their indicated ranges of n ,

$$\begin{aligned} a_{n,0} &= 1, \\ a_{n,1} &= n - 1, \\ a_{n,2} &= 4(n-2) + \binom{n-2}{2} = \frac{(n-2)(n+5)}{2}, \\ a_{n,3} &= 27(n-3) + 4(n-3)(n-4) + \binom{n-3}{3}. \end{aligned}$$

These give a combinatorial explanation of the initial columns of Table 2, rather than only an empirical pattern in the data.

12 Algorithms, experiments, and reproducibility

12.1 Two independent finite engines

Both engines represent a finite partial order by its inclusive principal upper sets as integer bit masks, and both validate reflexivity, antisymmetry, transitivity and mask bounds; the relation-based constructors first take a transitive closure and reject cycles.

The module `code/friendly.py` carries the constructions of Sections 3–4: the protected-root non-cut step of Lemma 3.4, the optimal certificate with its spanning-forest roots, the certificate verifier, lexicographic substitution, order duals, and the efficient ideal-based enumeration of naturally labelled orders. The module `code/finite_posets.py` is a separate implementation, with its own enumerator, its own residual dynamic program, its own search-based constructive-certificate routine, and the height, width, cover and product machinery used for the counterexample of Section 6. Agreement between them is agreement between two implementations, not a routine calling itself.

The direct algorithm, present in both, implements the defining residual equation:

```
def rank(mask):
    best = 0
    for x in elements(mask):
        if incomparable[x] & mask:
            after = mask & ~upper[x]
            best = max(best, 1 + rank(after))
    return best
```

Memoization makes each residual state run at most once. This algorithm *does not call* the component formula and does not use Lemma 3.4, so comparison against it is not a test of the formula against itself. It also reconstructs an optimal sequence and supplies an actual friend at every move, along with the residual before and after the move.

A second evaluator builds the incomparability graph, counts its components, and returns $n - c(P)$. On an explicit comparison matrix this needs $O(n^2)$ operations: forming the graph and searching its components are enough, and the multiset width is then reported symbolically as ω^k with no attempt to enumerate the infinite multiset order. If only covering or generating relations are supplied, a transitive closure comes first, and a naive transitivity check alone costs $O(n^3)$; the quadratic claim is for evaluation after a valid comparison matrix is given, not for every input representation. The direct rank algorithm has up to 2^n residual masks, so it is an exponential reference checker rather than the intended efficient method. Under unit-cost mask operations its bound is $O(n2^n)$ subset transitions.

For the constructive witness there are two routines, and their costs differ because their strategies differ. The routine in `code/friendly.py` implements Lemma 3.4 directly: one non-cut step selects a maximal vertex distinct from the protected root, computes the components after its deletion, and, if that vertex was a cut vertex, takes a maximal element of the last new component exactly as in the proof. Such a step costs $O(n^2)$ elementary operations on a comparison matrix, and there are at most $n - 1$ steps, so a full witness with its at most $n - 1$ choice–friend pairs is constructed in $O(n^3)$. The routine in `code/finite_posets.py` instead searches the maximal vertices and tests each candidate deletion, which is a conservative $O(n^4)$; it is kept because it is an independent implementation, not because it is the best bound available. Both are logically independent of the residual dynamic program. All these bounds are stated in the elementary-operation model: the Python implementations use arbitrary-precision integer bit sets, and we do not pretend that long machine integers have unit cost.

The verifier checks every emitted move against the original relation, not just the final length: it simulates the residuals, confirms that each named choice and friend is still present and incomparable, checks by union–find that the witness edges are acyclic, and checks that they span each original incomparability component with one declared root apiece. That establishes legality and confirms

the forest certificate of Proposition 4.2; optimality is then certified by comparing the sequence length with the component upper bound of Corollary 3.3.

12.2 Exhaustive finite checks

All partial orders compatible with the fixed linear extension $0 < 1 < \dots < n - 1$ were generated for $0 \leq n \leq 7$. These are *naturally labelled* posets. They are not all label permutations and are not quotiented by isomorphism. Every finite isomorphism class of the relevant size has at least one representative in this enumeration, generally several. To pass from size $n - 1$ to size n , the strict predecessor set of the new largest label is chosen among the order ideals of the previous order: every such ideal gives a valid natural order, and every natural order arises exactly once this way.

n	Total	$f = 0$	$f = 1$	$f = 2$	$f = 3$	$f = 4$	$f = 5$	$f = 6$
0	1	1	–	–	–	–	–	–
1	1	1	–	–	–	–	–	–
2	2	1	1	–	–	–	–	–
3	7	1	2	4	–	–	–	–
4	40	1	3	9	27	–	–	–
5	357	1	4	15	62	275	–	–
6	4,824	1	5	22	106	620	4,070	–
7	96,428	1	6	30	160	1,047	8,906	86,278

Table 2: Exhaustive naturally labelled orders by friendly rank, totalling 101,660 orders including the empty poset. For each order the test compares the direct residual rank with $n - c(P)$, repeats the direct recursion on the dual order, and constructs and simulates an optimal certificate with its spanning forest. The machine-readable form is `data/rank_distribution.csv`.

Every comparison passed. The maximal non-cut conclusion of Lemma 3.4 was additionally checked at the root of every connected case of size at least two, which is 90,655 explicit checks. Separately, the independent `FinitePoset` engine re-derived the same values, together with its own constructive certificates, on all 5,231 nonempty orders through size six and on the empty poset; this is a cross-implementation check over a smaller range, not a second exhaustive-enumeration figure.

The composition laws were tested as follows.

- *Products.* All 2,601 ordered pairs of naturally labelled factors of sizes zero through four, comparing the graph count and the independent residual recursion with (16), including the degenerate empty and singleton factors. Product supports reach sixteen elements. On the 2,401 of those pairs with both factors of size at least two, the unique non-endpoint incomparability component asserted by Theorem 5.2 was also checked.
- *Substitutions.* Every natural base of size at most four, with each fiber independently chosen from a singleton, a two-element chain, a two-element antichain and a three-element chain: 10,725 heterogeneous substitutions, each checked against the direct residual recursion and against (17).
- *Maximum friendly subsets.* Through support size five, 2,365 candidate subsets of the predicted maximum size were tested by a recursion restricting choices to that subset. Exactly the subsets described by Theorem 4.1 passed, and 1,052 corresponding prescribed-root certificates were constructed and verified.

- *Malformed input.* Eight malformed orders or malformed friend claims were presented and all eight were rejected.
- *Generating function.* The recurrence (30) and the convolution (31) reproduce every rank histogram in Table 2. This is a consistency check on the saved counts, not an independent enumeration.

The counterexample is verified separately, including height, width, endpoint flags, full relations, component lists, and optimal friendly sequences, as are the three-element separation of Example 6.1 and the extremal families of Proposition 6.3 for $3 \leq n \leq 7$. None of these results depends on trusting a random sample or a numerical ordinal approximation.

12.3 Finite checks of the corner strategy

The proof's finite successor-corner strategy has a separate simulation. For this purpose alone, each limit cutoff λ_i is replaced by the finite integer two. We use one, two, or three ordinal-coordinate slots, tail lengths from one to three, and all naturally labelled finite factors with at most three points. Single-chain cases are excluded exactly as in the theorem.

The resulting 387 cases all passed. In 385 cases the script checked that each prescribed friend was present and incomparable, that the number of selected points was N or $N - 1$ as appropriate, and that the final residual was exactly the simulated core. The remaining two cases have $N = 1$ and correspond to the proof's no-move subposet argument.

Replacing a limit by two does not verify limit saturation or core ranks. It tests only the finite move logic. The data file states this limitation explicitly. In particular these simulations are not evidence that finite replacement preserves transfinite friendly order type.

12.4 Hereditary Cantor normal forms

The module `code/ordinal_cnf.py` represents an ordinal below ε_0 as a finite decreasing tuple of pairs (exponent, positive coefficient), with each exponent itself represented in the same way. Zero is the empty tuple. This permits exact comparison, ordinary addition, natural sum, natural product, and right predecessor. A constructor rejects nondecreasing exponents or invalid coefficients.

The module also evaluates the proved ordinal-box and mixed-product formulas. This is a symbolic implementation of the theorem, not an independent oracle for the rank of an infinite tree. Twelve test ordinals were used for 7,200 sampled algebraic identity checks involving commutativity, associativity, and distributivity, in addition to finite arithmetic, successor, degeneracy, and worked-example checks. All six worked ordinal-box examples in the data file passed.

The exact mathematical theorem permits arbitrary ordinals, while this particular executable notation system does not encode ε_0 or ω_1 . The gap is a software representation choice, not a restriction hidden in the theorem.

12.5 Calculating a single finite example

The archive ships four concrete inputs under `examples/`, given by strict generating relations. For instance `examples/diamond.json` is

```
{"n":4,"relations":[[0,1],[0,2],[1,3],[2,3]]}
```

From the archive root, run

```
python3 code/friendly_cli.py examples/diamond.json --dp
python3 code/verify.py --max-n 7
```

The first command reports the incomparability components $\{0\}, \{1, 2\}, \{3\}$, friendly order type 1, multiset ordinal width $\omega^1(1)$, and an independently checked certificate. Input labels are integers from 0 through $n - 1$ and need *not* themselves form a linear extension; Hasse edges are acceptable, since the program takes the transitive closure and rejects cycles and out-of-range labels. With no "relations" key the input is an antichain. The optional key "roots":[...] prescribes exactly one root per incomparability component, each minimal inside its own component; the diamond, for example, accepts "roots":[0, 2, 3] and then selects only vertex 1. The `-dp` flag adds the exponential reference recursion and should be avoided on large inputs.

The pair `diamond.json` and `chain_plus_isolate.json` illustrates Example 6.1 one size up: both have four elements, height three and ordinary width two, yet friendly order types 1 and 3, hence multiset widths ω and ω^3 .

12.6 How to reproduce the results

Only Python 3.10 or later and its standard library are required for the computations. From the archive root, run

```
python3 code/verify.py
python3 code/check_distribution.py
```

The first script rewrites the files in `data/`:

```
verification_results.json
finite_verification.json, finite_verification.txt
rank_distribution.csv
generating_function_check.json
counterexample_certificate.json
ordinal_examples.json
examples_results.json
```

Use `-max-n` to change the exhaustive bound and `-out` to write the reports elsewhere; `python3 code/verify.py -max-n 4 -out /tmp/friendly-check` is a quick smoke test that still runs the composition and maximum-subset checks over their documented fixed ranges. The second script rebuilds the rank histograms from $A = 1/(1 - B)$ and compares them with the saved report. Tests use explicit checks rather than Python's optional `assert` statements, so running under `python -O` does not disable the verification conditions.

The delivered run completed in 40.685 seconds under Python 3.14.4 in this environment. That is a reproducibility datum, not a performance guarantee.

To rebuild the article, run `sh build.sh`; a standard LaTeX installation with the packages named in the source is required. The source uses New PX text and mathematics, and carries its bibliography inline, so no BibTeX pass is needed. No font files are distributed. The PDF in the archive has been compiled from the supplied source and visually checked. No proof assistant was used, and the archive contains no Lean code and no claim of machine-checked mathematical correctness.

13 Conclusions and remaining questions

The selected open direction admits a substantial exact treatment on a natural family of orders. Finite friendly order type is the number of vertices minus the number of incomparability components, equivalently the rank of the graphic matroid of the incomparability graph. That formula brings with it the classification of all maximum friendly subsets, explicit spanning-forest certificates, composition laws for ordinal sums, disjoint unions, Cartesian products and arbitrary nonempty finite lexicographic substitutions, sharp bounds, exact monotonicity under added comparisons, and an exact generating function for the distribution on naturally labelled posets. Together with the sum law it also evaluates every wpo whose incomparability components are all finite. Cartesian products collapse all non-endpoint components into one. For products of infinite ordinals and a finite poset, the infinite part is controlled by limit saturation and the finite correction by an explicit deletion of the upper corner. The final answer is the maximal order type itself, except for a right-predecessor correction when a greatest point exists.

The same analysis gives a clear obstruction to an overly small compositional interface. The four invariants (o, h, w, f) of the factors are insufficient for products, already in an eight-point example. In the finite case, cardinalities and endpoint flags repair the failure. The mixed theorem needs the finite cardinality, the greatest-element flag, and the ordinal factors themselves.

Several questions remain beyond the results proved here. For arbitrary wpos A, B , what additional structural information determines $f(A \times B)$? Can one describe a useful calculus for successor tails when the finite upper corner is replaced by a more complicated upper subposet? Can the finite deletion lemma and the mixed-product proof be formalized economically using existing libraries for finite posets, well-founded ranks, and ordinal natural arithmetic? These are research questions, not assumptions needed to validate the stated formulas.

The sharpest remaining direction is the friendly rank of *infinite connected* wpos, and how it responds to finite extensions. Two points delimit what the finite theory can contribute there. First, the finite construction is not a transfinite algorithm in disguise: it relies essentially on maximal vertices and on finite deletion, and an infinite wpo may have no maximal vertex at all. Second, the component decomposition (8) does reduce a general instance to its connected pieces plus ordinal addition, but it does not compute the ranks of those pieces. We offer no conjectural formula for them on the strength of the finite experiment alone, and Proposition 10.6 shows that no description of the abstract infinite incomparability graph can suffice unless the component order is retained along with it.

A particularly important caution concerns the role of upper sets. The finite upper-tail lemma works because a maximal extension of the core can be concatenated with the corner. A finite set that is not upper does not have this property, so arbitrary finite perturbations of wpos require a different analysis. Likewise, a general infinite factor may contain internal friendless regions rather than just endpoints; the product component theorem does not by itself control the rank after recursive residual deletions.

The results here therefore support a specific conclusion: the finite and mixed ordinal-product subproblems are solved by the supplied proofs, while the broad compositional research program remains larger than these families. The distinction between a proof and a novelty claim should be retained when circulating or developing this draft further.

A A small, inspectable certificate

Label (a_i, j) or (b_i, j) by the integer $2i + j$, for $j \in \{0, 1\}$. The inclusive upper sets, encoded as eight-bit masks, are

$$D \times K : (255, 170, 204, 136, 240, 160, 192, 128),$$

$$E \times K : (243, 162, 252, 168, 240, 160, 192, 128).$$

For example, bit y in mask u_x is set exactly when $x \leq y$. The following tables give optimal friendly sequences found by the direct residual algorithm. A friend may appear in more than one row, or be selected in a later row; it only has to be present at the indicated move.

Step	Selected point	Friend	Residual after the move
1	3	4	$\{0, 1, 2, 4, 5, 6\}$
2	5	2	$\{0, 1, 2, 4, 6\}$
3	6	1	$\{0, 1, 2, 4\}$
4	1	2	$\{0, 2, 4\}$
5	2	4	$\{0, 4\}$

Table 3: A length-five certificate for $D \times K$. Its original incomparability component sizes are 1, 6, 1, so a longer friendly sequence is impossible by Theorem 3.6.

Step	Selected point	Friend	Residual after the move
1	5	6	$\{0, 1, 2, 3, 4, 6\}$
2	1	2	$\{0, 2, 3, 4, 6\}$
3	6	3	$\{0, 2, 3, 4\}$
4	4	3	$\{0, 2, 3\}$
5	3	0	$\{0, 2\}$
6	0	2	$\{2\}$

Table 4: A length-six certificate for $E \times K$. Its original incomparability component sizes are 7, 1.

Each row can be checked directly by taking the preceding residual, checking the two incomparability relations, and deleting the upper set of the selected point. The first move in each table removes the greatest point without selecting it. The surviving residue is a chain, so no further friendly move is available. The JSON certificate additionally records each before-mask and after-mask, covers, endpoints, and invariant tuples.

B Proof and claim ledger

Claim	Justification	Status and scope
Definition of friendly order type	Vialard's invariant $o_{\perp}$ [1, Definition 3.3].	Imported; not introduced here.
Ordered components	Elementary order argument, Lemma 3.1.	Proved here; the argument is elementary and may well be folklore.
Protected-root non-cut lemma	Full finite proof, Lemma 3.4.	Proved here; the protected-root clauses are what Theorem 4.1 needs.
Finite formula $f(P) = P - c(P)$	Component-capacity upper bound plus constructive lower bound, Theorem 3.6.	Proved here for every finite poset; checked exhaustively through size seven in natural labellings, with a second engine through size six.
Maximum friendly subsets	Protected-root construction plus a necessity argument, Theorem 4.1.	Proved here; classifies the witness sets, not their orderings.
Forest certificates and graphic rank	Orientation by deletion time, Proposition 4.2 and Corollary 4.3.	Proved here for finite posets only; it fails on infinite ones by Proposition 10.6.
Product component theorem	Projection proof of the ordinal-cut lemma.	Proved here for arbitrary posets with two non-singleton factors; does not require wpo hypotheses.
Finite product formula, second proof	Chain-grid spanning subgraph and two incomparable anchors, Theorem 5.4.	Proved here; independent of the cut lemma, finite only. Retained deliberately alongside the projection route.
Substitution law	Fiber-lifting connectivity argument plus the finite formula, Theorem 5.6.	Proved here for arbitrary nonempty finite fibers; empty fibers need the stated preprocessing.
Disjoint-union law	Finite formula applied to a connected graph, Corollary 10.2.	Consistent with published finite formulas; not claimed as a new special case.
Counting generating function	Unique ordered component decomposition of natural labellings, Theorem 11.1.	Proved here; the numerical values a_n, b_n are outputs of the included enumeration.
Every ordinal realized	Transfinite sum law applied to two-element antichains, Corollary 10.5.	Proved here relative to the sum law.
Infinite graph-only obstruction	Isomorphic incomparability graphs with different ranks, Proposition 10.6.	Proved here; it bounds what any graph-only method can achieve.
Four-invariant obstruction	Explicit D, E, K with equal input tuples and outputs five and six.	Finite counterexample proved here; no exhaustive minimality claim.
Maximal linearizations and natural-product formula	de Jongh–Parikh [3].	Imported established background; not a new result.

Claim	Justification	Status and scope
Limit saturation	Vialard, Corollary 4.7 [1].	Imported; essential to the transfinite proof.
Both-limit product saturation	Vialard, Theorem 5.4.8(3) [2].	Already known; explicitly not claimed as new.
Mixed-product theorem	Exact core maximal order type, preservation of friends in the core, explicit corner witnesses.	Proved here relative to the imported background, for arbitrary infinite ordinal factors and a finite nonempty factor, with the stated chain exception.
Multiset-width consequences	Substitute the computed f into Vialard's Theorem 3.4.	Corollaries of an imported theorem, not a new multiset-width theorem.
Binary ordinal sum	Known identity, with a proof included.	Not a novelty claim. The ordinal-indexed extension is proved in this report.
Computer verification	Independent finite residual DP, constructive sequences, corner simulations, symbolic arithmetic tests.	Finite and symbolic checks only; no Lean formalization or infinite rank computation.
Novelty	Source check described in Section 1.	Exact formulas not located in the checked sources; no priority certification or peer review.

Checks that prevent common misstatements

The following conditions are easy to drop and each of them matters.

- The friend must be present in the *current residual*, not merely somewhere in the original poset.
- Isolated vertices of $\text{Inc}(P)$ count as components. The empty poset has $c = 0$ and $f = 0$, and its multiset order has width $1 = \omega^0$, not 0.
- Protected roots must be minimal *inside their own component*, not necessarily minimal in the whole poset.
- A forest root belonging to a later component may be removed while an earlier component is being processed. It remains a root of the certificate forest without remaining an element of the final residual.
- Empty substitution fibers require deleting their base vertices and taking the induced base order first; they cannot be fed to (17) directly.
- The product formula (16) requires both factors to have at least two elements. Empty and singleton factors are separate cases.
- The endpoint flags record *global* least and greatest elements, not minimal and maximal ones.
- The outer addition in a transfinite sum is ordinal addition, not natural addition and not cardinal addition, and it is taken in the displayed order.
- The finite component upper bound distinguishes points that are *selected* from points removed incidentally.

- The product cut of Lemma 5.1 requires every point of L below every point of U , not merely that L be a downset.
- The finite upper-tail Lemma 7.3 needs T upper and $|T|$ equal to the final finite Cantor coefficient.
- The mixed-product theorem excludes the single-chain case, and its $N = 1$ corner uses subposet monotonicity without any claimed legal corner selection.
- The right predecessor must not be confused with left ordinal subtraction.
- The hereditary Cantor normal form implementation is restricted to ordinals below ε_0 ; the theorems are not.
- Set-sized posets and ordinary ZFC are assumed throughout.

C A short executable ordinal example

From the archive root, the following code computes the right predecessor of the natural product in (20) using the supplied module.

```
import sys
sys.path.insert(0, "code")
from ordinal_cnf import Ord, OMEGA, ordinal_box_friendly

alpha = OMEGA.add(Ord.nat(2))
beta = OMEGA.add(Ord.nat(3))
value = ordinal_box_friendly([alpha, beta])
print(value)
#  $\omega^{(2)} + \omega * 5 + 5$ 
```

For a finite factor without a greatest point, use the mixed evaluator:

```
from ordinal_cnf import mixed_product_friendly

alpha = OMEGA.add(Ord.nat(1))
value = mixed_product_friendly(
    [alpha], cardinality=2, has_greatest=False
)
print(value)
#  $\omega * 2 + 2$ 
```

The input flag in the second example represents a two-point antichain. The module cannot infer this flag from a mere cardinality. This is a small executable reflection of the mathematical distinction exposed by the finite counterexample.

References

- [1] Isa Vialard. Ordinal measures of the set of finite multisets. In *48th International Symposium on Mathematical Foundations of Computer Science (MFCS 2023)*, Leibniz International Proceedings in Informatics, vol. 272, article 87, pp. 87:1–87:15, 2023. DOI: 10.4230/LIPIcs.MFCS.2023.87. Official proceedings version: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2023.87>.
- [2] Isa Vialard. *Measuring well quasi-orders and complexity of verification*. PhD thesis, Université Paris-Saclay, 2024. Author-hosted version consulted, with Section 5.4 on pp. 74–76 and the conclusion beginning on p. 109: <https://isavialard.github.io/home/mwqo.pdf>. Different hosted versions have different pagination; theorem numbers and the cited author-hosted version identify the results used here.
- [3] D. H. J. de Jongh and R. Parikh. Well-partial orderings and hierarchies. *Indagationes Mathematicae*, 39(3):195–207, 1977. DOI: 10.1016/1385-7258(77)90067-1.
- [4] Isa Vialard. Ordinal measures of the set of finite multisets. arXiv:2302.09881, version 1, 20 February 2023. <https://arxiv.org/abs/2302.09881>. Listed to disambiguate the earlier terminology “maximal safe order type”; the present report follows the published version instead.
- [5] Isa Vialard. Author website and publication list. <https://isavialard.github.io/home/>. Consulted 19 September 2026 for the limited source-status check.
- [6] Mirna Džamonja, Sylvain Schmitz, and Philippe Schnoebelen. On ordinal invariants in well quasi orders and finite antichain orders. In *Well Quasi-Orders in Computation, Logic, Language and Reasoning*, edited by Peter Schuster, Monika Seisenberger, and Andreas Weiermann, Trends in Logic, vol. 53, pp. 29–54. Springer, 2020. DOI: 10.1007/978-3-030-30229-0_2. Cited for the standard rank conventions for ordinal invariants of well-quasi-orders. See also the authors’ revised preprint, arXiv:1711.00428v4, including its erratum: <https://arxiv.org/abs/1711.00428>.